

KENDRIYA VIDYALAYA SANGATHAN, RAIPUR REGION**FIRST PRE-BOARD EXAM (2024-25)****SUBJECT: MATHEMATICS (BASIC)**

CLASS: X

MM : 80 MARKS

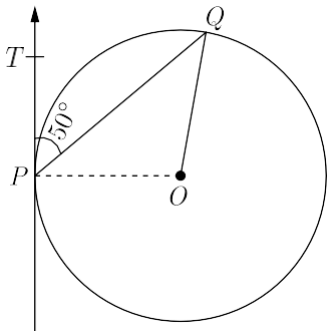
TIME : 3 HOURS

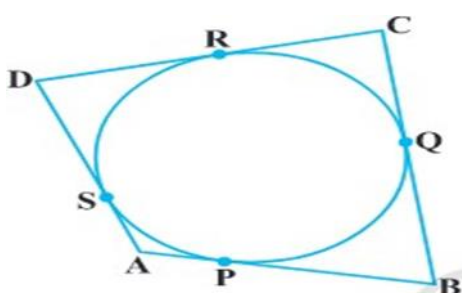
General Instructions:

1. This Question Paper has 5 Sections A-E.
2. Section A has 20 MCQs carrying 1 mark each
3. Section B has 5 questions carrying 02 marks each.
4. Section C has 6 questions carrying 03 marks each.
5. Section D has 4 questions carrying 05 marks each.
6. Section E has 3 case based integrated units of assessment (04 marks each) with subparts of the values of 1, 1 and 2 marks each respectively.
7. All Questions are compulsory. However, an internal choice in 2 Qs of 5 marks, 2 Qs of 3 marks and 2 Questions of 2 marks has been provided

SECTION A

Q.1	If $\frac{4}{5}$, a, 2 are three consecutive terms of an A.P., Then find the value of a: (a) $\frac{5}{7}$ (b) $\frac{5}{4}$ (c) $\frac{1}{5}$ (d) $\frac{7}{5}$	(1)
Q.2	If α and $1/\alpha$ are the zeroes of the quadratic polynomial $2x^2 - x + 8k$, then k is a) 4 b) $1/4$ c) $-1/4$ d) 2	(1)
Q.3	The value of p for which the system of equations $6x + 5y = 4$ and $12x + py = -8$, has no solution is _____ (a) 10 (b) -10 (c) $5/2$ (d) $2/5$	(1)
Q.4	If $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$, then the lines represented by the equations $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ are _____ (a) Intersecting lines (b) coincident lines (c) Parallel lines (d) none of these	(1)
Q.5	The value of x in $x - \frac{81}{x} =$ is : (a) real and unequal (b) real and equal (c) roots are imaginary (d) roots are not equal	(1)
Q.6	D and E are the midpoints of side AB and AC of a triangle ABC, respectively and BC = 6 cm. If $DE \parallel BC$, then the length (in cm) of DE is: (a) 2.5 (b) 3 (c) 5 (d) 6	(1)
Q.7	If ABC and DEF are two triangles and $AB/DE = BC/FD$, then the two triangles are similar if : (a) $\angle A = \angle F$ (b) $\angle B = \angle D$ (c) $\angle A = \angle D$ (d) $\angle B = \angle E$	(1)
Q.8	The point on the X axis which is equidistant from (-1,0) and (5,0) is (a) (0,2) (b) (2,0) (c) (3,0) (d) (0,3)	(1)
Q.9	The mid point of the segment AB is the point P(0,4). If the coordinates of B are (-2,3), then the coordinates of A are : (a) (2,5) (b) (-2,-5) (c) (2,9) (d) (-2,11)	(1)

Q.10	A tree casts a shadow 15 m long on the level of ground, when the angle of elevation of the sun is 45° . The height of a tree is (a) 10 m (b) 14 (c) 8 m (d) 15 m	(1)
Q.11	$(\sin 30^\circ + \cos 60^\circ) - (\sin 60^\circ + \cos 30^\circ)$ is equal to: (a) 0 (b) $1+2\sqrt{3}$ (c) $1-\sqrt{3}$ (d) $1+\sqrt{3}$	(1)
Q.12	In figure, O is the centre of circle. PQ is a chord and PT is tangent at P which makes an angle of 50° with PQ , then $\angle POQ$ is  (a) 130° (b) 90° (c) 100° (d) 75°	(1)
Q.13	If the sum of the areas of two circles with radii R_1 and R_2 is equal to the area of a circle of radius R , then (A) $R_1 + R_2 = R$ (B) $R_1^2 + R_2^2 = R^2$ (C) $R_1 + R_2 < R$ (D) $R_1^2 + R_2^2 < R^2$	(1)
Q.14	The base radii of a cone and a cylinder are equal. If their curved surface areas are also equal, then the ratio of the slant height of the cone to the height of the cylinder is (a) 2:1 (b) 1 : 2 (c) 1 : 3 (d) 3 : 1	(1)
Q.15	If r is the radius of the sphere, then the surface area of the sphere is given by; (a) $4 \pi r^2$ (b) $2 \pi r^2$ (c) πr^2 (d) $4/3 \pi r^2$	(1)
Q.16	In a frequency distribution, the mid value of a class is 10 and the width of the class is 6. The lower limit of the class is (a) 6 (b) 7 (c) 8 (d) 12	(1)
Q.17	If a card is selected from a deck of 52 cards, then the probability of its being a red face card is (a) $1/13$ (b) $2/13$ (c) $1/2$ (d) $3/26$	(1)
Q.18	Two coins are thrown at the same time. Find the probability of getting both heads. (a) $3/4$ (b) $1/4$ (c) $1/2$ (d) 0	(1)
	DIRECTION: In the question number 19 and 20 a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A) (b) Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A) (c) Assertion (A) is true but reason (R) is false. (d) Assertion (A) is false but reason (R) is true.	

Q.19.	Assertion(A): The sequence $-1, -1 - 1, \dots, -1$ is an AP. Reason(R): In an AP, $a_n - a_{n-1}$ is constant where $n \geq 2$ and $n \in \mathbb{N}$	(1)
Q.20	Assertion(A): $(2 + \sqrt{3})\sqrt{3}$ is an irrational number. Reason(R): Product of two irrational numbers is always irrational.	(1)
	SECTION B	
Q.21	Find the smallest number which is divisible by both 306 and 657.	(2)
Q.22	For what value of k is -4 a zero of the polynomial $f(x) = x^2 - x - (2k+2)$ OR If α, β are zeroes of the quadratic polynomial $2x^2 + 7x + 5$, then find the value of $\alpha^2 + \beta^2 + \alpha\beta$	(2) (2)
Q.23	$ABCD$ is a trapezium in which $AB \parallel CD$ and its diagonals intersect each other at the point O . Show that $AO/BO = CO/DO$	(2)
Q.24	Find the value of $\cos 2\theta$, if $2 \sin 2\theta = \sqrt{3}$. OR If $(A - B) = 1/2$ and $\cos(A + B) = 1/2$, $0^\circ < A + B < 90^\circ$ and $A > B$, then find the values of A and B .	(2) (2)
Q.25	From a point Q , the length of the tangent to a circle is 24 cm and the distance of Q from the centre is 25 cm. Find the radius of the circle.	(2)
	SECTION C	
Q.26	The sum of a two-digit number and the number obtained by reversing the order of its digits is 99. If ten's digit is 3 more than the unit's digit, then find the number	(3)
Q.27	Prove that $\sqrt{3}$ is an irrational number.	(3)
Q.28	A quadrilateral $ABCD$ is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$  OR Prove that the lengths of tangents drawn from an external point to a circle are equal.	(3) (3)
Q.29	2 cubes each of volume 64 cm^3 are joined end to end. Find the surface area of the resulting cuboid.	(3)
Q.30	A bag contains a red ball, a blue ball and a yellow ball, all the balls being of the same size. Kritika takes out a ball from the bag without looking into it. What is the probability that she takes out the (i) yellow ball? (ii) red ball? (iii) blue ball?	(3)

Q.31	Prove that: $(\operatorname{cosec} A - \sin A)(\sec A - \cos A) = \frac{1}{\tan A + \cot A}$	(3)																						
	SECTION D																							
Q.32	A train travels 360km at a uniform speed. If the speed had been 5km/h more, it would have taken 1 hour less for the same journey. Find the speed of the train. OR A motor boat whose speed is 18km/h in still water takes 1 hour more to go 24km upstream than to return downstream to the same spot. Find the speed of the stream.	(5)																						
Q.33	If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that the other two sides are divided in the same ratio.	(5)																						
Q.34	The person standing on the bank of river observes that the angle of elevation of the top of a tree standing on opposite bank is 60°. When he moves 30 m away from the bank, he finds the angle of elevation to be 30°. Find the height of tree and width of the river. OR Poles equal height are standing opposite to each other on either side of the road which is 80m wide. From a point between on the road the angles of elevation of the tops of the poles are 60° and 30° respectively. Find the height of poles and the distance of the points from poles.	(5)																						
Q.35	The median of the following data is 525. Find the values of x and y, if the total frequency is 100. <table><tr><th>Class Interval</th><th>Frequency</th></tr><tr><td>0-100</td><td>2</td></tr><tr><td>100-200</td><td>5</td></tr><tr><td>200-300</td><td>X</td></tr><tr><td>300-400</td><td>12</td></tr><tr><td>400-500</td><td>17</td></tr><tr><td>500-600</td><td>20</td></tr><tr><td>600-700</td><td>Y</td></tr><tr><td>700-800</td><td>9</td></tr><tr><td>800-900</td><td>7</td></tr><tr><td>900-1000</td><td>4</td></tr></table>	Class Interval	Frequency	0-100	2	100-200	5	200-300	X	300-400	12	400-500	17	500-600	20	600-700	Y	700-800	9	800-900	7	900-1000	4	(5)
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	SECTION E (Case Study Based questions)																							
Q.36	India is competitive manufacturing location due to the low cost of manpower and strong technical and engineering capabilities contributing to higher quality production runs. The production of TV sets in a factory increases uniformly by a fixed number every year. It produced 16000 sets in 6th year and 22600 in 9th year.	(4)																						



Based on the above information, answer the following questions:

- (i) Find the production during first year.
- (ii) Find the production during 8th year.
- (iii)(a) Find the production during first 3 years.

OR

- (b) In which year, the production is Rs 29,200

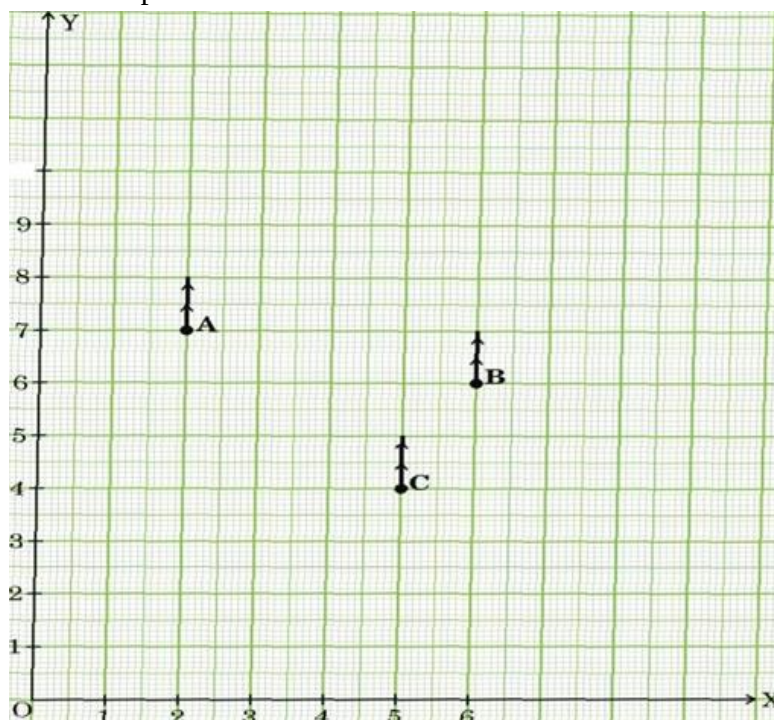
Q.37 Resident Welfare Association (RWA) of Gulmohar Society in Delhi, have installed three electric poles A, B and C in the society's common park. Despite these three poles, some parts of the park are still in the dark. So, RWA decides to have one more electric pole D in be modelled as a coordinate system given below. (4)

On the basis of the above information, answer the following questions :

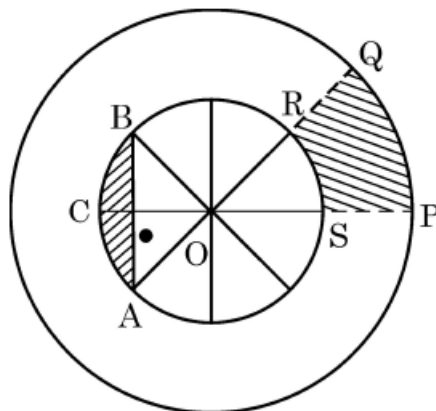
- (i) What is the position (coordinates) of the pole C ?
- (ii) What is the distance of the pole B from the corner O of the park ?
- (iii)(a) Find the position of the fourth pole D so that the four points A, B, C and D form a parallelogram ABCD.

OR

- (b) Find the distance between poles A and C.



- Q.38 NSS (National Service Scheme) aims to connect the students to the community and to involve them in problem solving process. NSS symbol is based on the 'Rath' wheel of the Konark Sun Temple situated in Odisha. The wheel signifies the progress cycle of life. The diagrammatic representation of the symbol is given below :



Observe the figure given above. The diameters of inner circle are equally placed. Given that $OP = 21$ cm, $OS = 10$ cm.

Based on the above information, answer the following questions :

- (i) Find $\angle ROS$.
- (ii) Find the perimeter of sector OPQ .
- (iii) (a) Find the area of shaded region $PQRS$.

OR

- (iii) (b) Find the area of shaded region ACB i.e. the segment ACB .